

Real Estate Market Place

Project Scope Statement

1. Project Description

This is a comprehensive online real estate marketplace designed to connect multiple stakeholders in the property ecosystem, including **buyers, sellers, agents, lenders, developers, and advertisers.**

The platform will provide:

- A **Client Web & Mobile Application** for property discovery, comparisons, collaboration, and transactions.
- An **Agent Web & Mobile Application** for lead management, listings, marketing, and communication.
- An **Admin Web Panel** for managing users, listings, transactions, promotions, and the affiliate system.

The system will support multi-language accessibility, seamless payment processing, integration with communication and collaboration tools, and an automated affiliate/referral system.

2. System Users

The system will include the following user types:

- **Admin Users**
 - Super Admin (full control).
 - Sub Admins (role-based access e.g., finance, support).

- **Clients (Buyers/Sellers)**
 - Search, compare, and save properties.
 - Communicate with agents.
 - Manage transactions and documents.
- **Real Estate Agents**
 - Independent agents.
 - Agency agents (individual or franchise).
 - Manage listings, clients, leads, and performance.
- **Lenders**
 - Banks.
 - Microfinance institutions.
 - Pawnshops.
 - Provide financing options and loan calculators.
- **Real Estate Developers**
 - Flat & villa developers.
 - Condo/apartment developers.
 - Manage large-scale projects and listings.
- **Advertisers**
 - Promote services/products on the platform.

- **Property Management Firms / Title Companies / Lawyers / Appraisers**
 - Provide additional legal and property management services.

3. Localization & Language Support

- Languages Supported:
 - English.
 - Khmer.
 - Chinese.
- Fallback Translation: Google Translate integration (for other languages).

4. Third Party Integrations

Functionality	Selected/Preferred Option	Alternatives / Notes
CRM Integration	Bitrix	Open for other CRM suggestions
Communication	Telegram	Primary platform for client-agent interactions
E-Signature	DocuSign	Industry-standard solution

E-Document (Collaboration)	Open for suggestions	Must support editing + collaboration features
Meeting Scheduling	Calendly with Google Calendar	One of these will be implemented
Maps & Location	Google Maps	For property pinning, boundaries, and navigation
Promo Code & Discount Engine	Open for suggestions	Needs admin control + reporting
Hosting	AWS (Amazon Web Services)	Scalable cloud infrastructure
Mobile App Framework	Flutter (Cross-Platform)	Single codebase for iOS and Android

5. **MLS (Multiple Listing Service)**

- No external MLS integration.
- Properties will be **added manually** by agents, developers, or admins.

6. **Not in Scope for MVP**

- AI-driven features.
- Data Migration.
- Video Hosting

Now let's start with each user of the system. We will describe each module/functionality in detail and also specify their flow from one module to another.

1. Super Admin (WebPanel)

The **Admin Module** is the **control center** of the platform. It enables **Super Admins** and **Sub Admins** to manage:

- Users (clients, agents, affiliates, lenders, developers, advertisers).
- Property listings and approvals.
- Transactions, commissions, and payouts.
- Promotions, discounts, and referral systems.
- Reports, analytics, and platform configurations.
- Security, authentication, and activity tracking.

1.1 Authentication

Login Page (Frontend Requirements)

Fields:

1. Email/Username

- Input type: `text`
- Validation:
 - Must be in email format (`name@domain.com`) if email.
 - Username is allowed only if the system supports it.
 - Required (cannot be blank).

2. Password

- Input type: `password`

- Validation:
 - Minimum 8 characters.
 - Must include at least 1 uppercase, 1 lowercase, 1 number, 1 special character.
 - Required (cannot be blank).

3. Remember Me (optional)

- Input type: **checkbox**
- Function: Store session token in local storage/cookies if checked.

4. Login Button

- Triggers authentication request to backend API.

5. Forgot Password Link

- Redirects to password reset flow.
- **Forgot Password** → Navigates to Password Reset.

Password Reset Flow

Trigger: Forgot Password

- UI: On the login page, Admin clicks “Forgot Password?”.
- Field: Email (required).
- Validation:
 - Must be a valid email format.
 - Must exist in **admin_users** table.
- Action: Send reset link to registered email.

Forgot Password API

- POST `/api/admin/forgot-password`

Reset Password Page (Frontend)

- Fields:
 - New Password (with strength validation).
 - Confirm Password (must match new password).
- Validation Rules:
 - Minimum 8 chars.
 - At least 1 uppercase, 1 lowercase, 1 digit, 1 special char.
- Button: Submit → calls reset API.

Reset Password API

- POST `/api/admin/reset-password`

Navigation: On success → “Password changed, please login” → redirects to **Login Page**.

Backend Authentication Flow

Login API Endpoint:

- POST `/api/admin/login`

Two-Factor Authentication (2FA)

- After successful login (valid email/password), the system checks if 2FA is enabled.
- Options for 2FA:
 - Email OTP → sent to admin's registered email.
 - SMS OTP → sent to registered phone (if phone linked).
 - Authenticator App → Google Authenticator / Authy.

2FA API Endpoint:

- POST `/api/admin/verify-2fa`

Session Management

- Token-based Authentication (JWT):
 - `accessToken` (short-lived, e.g., 15–60 mins).
 - `refreshToken` (long-lived, e.g., 7–30 days).
- Auto Logout after inactivity (configurable, e.g., 30 mins).
- “Remember Me” extends the session by using `refreshToken`.

Navigation: Redirect to Admin Dashboard upon successful login.

1.2 Admin Dashboard (Landing Page)

Purpose

The **Admin Dashboard** is the **first screen** an admin sees after login. It gives a **snapshot of platform activity** (KPIs, widgets, quick actions) and enables navigation to other modules.

KPIs & Widgets (Top Section)

Each KPI is a **card widget** with: Title, Value.

1. Total Users (by type)

- Clients
- Agents
- Lenders
- Developers
- Advertisers

2. Active Listings (by category)

- For Sale
- For Rent
- Foreclosures / Bank-owned

3. Sales Revenue

- Daily, Weekly, Monthly, Year-to-Date
- Chart view (line or bar chart).

4. Referral Activity

- Total affiliates.
- Sales generated through referrals.

5. Active Promotions

- Count of running promotions.
- Number of redemptions so far.

Quick Actions:

- Add/Edit Promotion.
- View Pending Agent Approvals.
- View Pending Affiliate Approvals.
- Access Reports.

Navigation: Clicking a widget navigates to its module

(e.g., “Agents” → Agent Management).

Now we look into each user and their role management in the admin panel. Clicking on any widget it will redirect to the respective module. Lets check these modules/users in detail.

1.3 User Management

The User Management section allows the Admin to:

- Manage all users of the system (Clients, Agents, Lenders, Developers, Advertisers, Affiliates, Admins).
- Create, edit, suspend, or delete accounts.

- Track user activity and maintain compliance logs.

Navigation: From Dashboard → “Users” → User Management.

User Types: These are the types of users in the system

- Clients.
- Agents.
- Lenders.
- Developers.
- Advertisers.
- Affiliates.

Features/Functionality

User List Page

- View Users → List with filters (name, email, role, status).

Purpose: View and manage all registered users.

UI Components:

- **Table/List of Users** with columns:
 - User ID (system-generated, unique).
 - Full Name.

- Email.
- Phone Number.
- User Type (Client, Agent, Lender, etc.).
- Role (for Admin users only).
- Status (Active, Suspended, Pending Approval).
- Created Date.
- Last Login.
- Actions (View, Edit, Suspend, Delete).

- **Filters (left panel or top bar):**

Purpose: To sort users in desired format.

- By User Type.
- By Status.
- By Date Created.
- By Last Login.

- **Global Search Bar:** Search by name, email, or phone.

- **Buttons:**

-  Add New User - This will allow the admin to add/create a new user in the system.

Validations:

- Search must match valid string/email/phone format.

Navigation: Clicking on a user → **User Detail Page.**

Add New User

Purpose: Manually create a user (useful for Admin onboarding or special cases).

Fields:

- Full Name (required, min 3 chars).
- Email (required, must be unique, valid email format).
- Phone Number (optional, but must be valid format if entered).
- Password (required, strong password rules).
- Confirm Password (must match Password).
- User Type (dropdown: Client, Agent, Lender, Developer, Advertiser, Affiliate, Admin).
- Status (Active, Pending, Suspended).
- Role (dropdown → only visible if User Type = Admin).

Buttons:

-  Save.
-  Cancel (back to User List).

Validations:

- All required fields must be filled.
- Email must be unique in the DB.
- Password + Confirm Password must match.

Navigation: On Save → User List refreshed with **new user**.

User Detail Page

When an Admin clicks on a user from the User List, they land on the User Detail Page.

Purpose: View and manage a single user's profile & activity.

Sections: It shows what information should be included in each section of the user detail page.

1. Basic Info

- Name, Email, Phone, Type, Status.
- Edit button → opens Edit User form.

2. Account Actions (Buttons):

- Suspend (confirm popup: "Are you sure?").
- Delete (permanent, with warning popup).
- Reset Password (sends email with reset link).
- Resend Verification Email (if pending).

3. User Activity Logs:

- Last Login Date & IP.
- Number of logins in last 30 days.
- Actions performed (listings posted, purchases, reviews, etc.).

4. Linked Data in user detail page (Tabs):

→ **Properties tab** (if the user is Agent/Developer).

Clicking on the tab it should show the property listed by the respective user.

Tabs → Show filtered data without leaving the page.

Fields shown:

- Property ID.
- Title.
- Type (Sale, Rent, Foreclosure).
- Status (Pending, Approved, Rejected, Active, Expired).
- Location (Google Map Pin).
- Created Date.

Actions (per property):

- View Property.
- Approve/Reject (if pending).
- Edit (price, description, photos).
- Delete.
- Mark as Featured.
- Admin can **approve, reject, edit, delete, or feature** properties directly.

Navigation: Clicking property → Property Detail Page (with full info).

→ **Transactions tab** (if Client/Agent).

Displays all financial transactions the user is part of (buying, selling, renting).

- Useful for tracking payment history, commissions, and disputes.
- Admin can mark transactions as reviewed or flag them for investigation.

Fields shown:

- Transaction ID.
- Property Title/ID.
- Buyer/Seller (if applicable).
- Amount.
- Commission.
- Date.
- Status (Completed, Pending, Disputed).

Actions (per transaction):

- View Details.
- Mark as Reviewed.
- Flag Dispute.

Navigation: Clicking transaction → Transaction Detail Page.

→ **Referrals tab** (if Affiliate)

Shows affiliate and referral activity for the user.

- Displays direct signups and downline affiliates (chain referrals).
- Helps Admin monitor sales generated and commissions earned.
- Provides control for **approving or rejecting commission payouts**.

Fields shown:

- Referral ID.
- Referral Code.
- Direct Referrals (list of users signed up).
- Downline Referrals (2nd level).
- Sales Attributed.
- Commission Earned.
- Payout Status.

Actions:

- View Referral Chain (tree view: Affiliate → Direct → Downline).
- Approve/Reject pending commissions.

Navigation: Clicking referral → Referral Detail Page.

→ **Reviews & Ratings Tab (For Agents/Developers)**

Shows **feedback received** from clients.

- Admin can see **ratings and written reviews** about an agent/developer.
- Helps ensure **quality control** and remove **fake/abusive reviews**.

Fields shown:

- Review ID.
- Client Name (who submitted review).
- Rating (1–5 stars).
- Review Comment.
- Date.

Actions:

- Remove Review (to filter fake/abusive reviews).
- Flag Review.

Navigation: Clicking review (View more) → Review Detail Modal.

→ Documents Tab (For Agents/Developers/Advertisers)

Contains all uploaded documents by the user (e.g., licenses, contracts, ID proofs).

- Allows Admin to verify authenticity of professional/legal documents.
- Provides options to approve, reject, or download documents.

Fields shown:

- Document ID.
- Document Type (License, ID Proof, Title Deed, Contract, Marketing Content).
- Upload Date.
- Verification Status (Verified, Pending, Rejected).

Actions:

- View Document.
- Approve/Reject Document.
- Download.

Navigation: Clicking document → Opens document preview.

Navigation Buttons:

- **Back button** → User List.
- **Tabs** → Show filtered data without leaving the page.

User-Type vs Linked Data Matrix

User Type	Properties Tab	Transactions Tab	Referrals Tab	Tickets Tab	Reviews & Ratings Tab	Documents Tab
Client (Buyer/Seller)	✗	✓	✗	✓	✗	✗
Agent (Independent/ Agency)	✓	✓	✗	✓	✓	✓
Developer (Flat/Condo Builder)	✓	✓	✗	✓	✓	✓
Lender (Bank/Microfinance/ Pawnshop)	✗	✓	✗	✓	✗	✓
Advertiser	✗	✗	✗	✓	✗	✓
Affiliate	✗	✗	✓	✓	✗	✗
Admin (Super/Sub)	✗	✗	✗	✓	✗	✗

Edit User (Form)

From the list of users admin can find some actions for each user.

Actions: Delete, Edit , View

Navigation: List of Users -> Select a User -> Click on Action **Edit**.

Fields (pre-filled with existing data):

- Full Name.
- Email (read-only).
- Phone Number.
- User Type (dropdown, cannot be changed once set).
- Status (Active, Suspended, Pending).
- Role (dropdown → for Admins only).

Buttons:

-  Save Changes - It should save the changes made
-  Cancel - It should not save the changes and return to the list of users.

Validations:

- Full Name cannot be blank.
- Email cannot be changed.
- Status must be a valid enum.

Navigation: On Save → Redirect to User Detail Page.

Bulk Actions

Select multiple users via checkboxes in the User List.

- Bulk Suspend.
- Bulk Activate.
- Bulk Export.

Navigation: Confirmation popup before applying action.

Security & Validation

- Unique constraints: Email, User ID.
- Audit logs: Record every Add/Edit/Delete action by admin.
- Only Super Admin can delete Admin accounts.

1.4 Property Listings Management

The **Property Listings Management** module allows Admins to **monitor, review, approve, and manage** all property listings submitted by **Agents, Developers, or Admins**.

This ensures that only **valid, verified, and high-quality listings** are displayed to clients on the platform. Admins can also **edit, feature, or remove** listings when necessary.

Navigation to Property Listings Management

1. Login → Dashboard

- After logging in, Admin lands on the **Dashboard**.

2. Access via Navigation Menu (Sidebar/Topbar):

- Click “**Listings**” in the Admin menu.
- Redirected to **Listings Dashboard**.

3. Access via Dashboard Widget:

- On the Dashboard, under **KPIs**, Admin sees **Active Listings** card (with counts for Sale, Rent, Foreclosure).
- Clicking the KPI card → redirects to **Listings Dashboard (filtered view)**.

4. Admin clicks Property → goes to Property Detail Page.

5. Admin **approves, rejects, edits, or deletes** listing.

6. Admin can **bulk manage** listings directly from the dashboard.

Listings Dashboard

Purpose: Overview/List of all property listings.

UI Components (Table View):

- Columns:
 - Property ID (auto-generated).
 - Title.
 - Type (Sale, Rent, Foreclosure).
 - Category (Flat, Villa, Condo, Land, Commercial).
 - Location (City, District, Google Map Pin).
 - Agent/Developer Name.
 - Status (Pending, Approved, Rejected, Active, Expired).
 - Date Created.
 - Last Updated.
 - Actions (View, Approve, Reject, Edit, Delete, Feature).

Filters (Top Panel):

- Property Type (Sale, Rent, Foreclosure).
- Property Category.
- Status.
- Agent/Developer.
- Location.
- Date Range.

Buttons: Add these button on the top panel

-  Add Listing (if Admin wants to manually create).
-  Export Listings (CSV, Excel, PDF).

Navigation:

- Click Property → Property Detail Page.
- Click “Approve”/“Reject” → opens confirmation popup.

Property Detail Page

Purpose: View and verify full details of a property before approval.

Sections:

1. **Basic Info:** Title, Description, Type, Category, Price, Status.
2. **Location Info:** Address, Google Map Pin.
3. **Media:** Image gallery, floor plans(property dimensions), documents.
4. **Linked User:** Agent/Developer who submitted it.
5. **Listing History:** Creation date, edits, approvals.

Admin Action Buttons:

- Approve Listing → Makes property visible to clients.
- Reject Listing → Requires reason (text field).
- Edit Listing → Opens editable form (price, description, media).

- Delete Listing → Permanent removal.
- Feature Listing → Pushes listing to top in search results.

Validations:

- If Reject → “Reason” field is required.
- If Edit → All required fields (title, price, location) must be valid.

Navigation:

- **Approve/Reject** → Returns to Listings Dashboard (with success message).
- **Edit** → Opens Edit Listing Form.

Add/Edit Listing (Form)

Edit property form opens with the prefilled data.

Using the Add button on the top panel on Listings dashboard, admin can add a new listing.

Fields (with validation):

- Title (required, min 5 chars).
- Description (required, min 20 chars).
- Property Type (dropdown: Sale, Rent, Foreclosure).
- Category (dropdown: Flat, Villa, Condo, Land, Commercial).
- Price (required, numeric, >0).
- Location (required, must integrate with Google Maps API).
- Upload Images (at least 1 required, max 10).
- Upload Documents (optional, e.g., ownership proof).
- Status (default = Pending for new).

Buttons: Show on both add and edit form.

-  Save.
-  Cancel (returns to Listings Dashboard).

Navigation: Save → Redirects to Property Detail Page with success message.

Bulk Actions (on Listings Dashboard)

- Select multiple listings via checkboxes.
- Bulk Approve.
- Bulk Reject (requires reason).
- Bulk Delete.
- Bulk Export.

Navigation: Confirmation popup for each bulk action.

Reject Listing Flow (Admin → Property Management)

Purpose

The **Reject Listing** flow allows Admins to **decline a property listing** submitted by an Agent/Developer if it is **invalid, incomplete, fraudulent, or against policy**.

This flow ensures:

- Proper **moderation of content**.
- Clear **communication to the submitting user**.
- **Tracking of rejection reasons** for audits & transparency.

→ Process:

On the **Listings Dashboard**, each row (listing) has an **Actions column**.

Actions available: **View | Approve | Reject | Edit | Delete | Feature**.

- When Admin clicks **Reject**, a **confirmation modal popup** opens.

Reject Listing Modal (Popup)

Fields:

- **Title:** “Reject Listing – [Property Title]”
- **Reason for Rejection (Required)**
 - Dropdown with common reasons (configurable by Admin):
 - Missing/Incorrect Property Info.
 - Invalid Price/Details.
 - Low-Quality Images.
 - Duplicate Listing.
 - Fraudulent or Suspicious.

- Other (requires custom text input).
- **Additional Notes (Optional)** – free text area for detailed explanation.

Buttons:

-  Confirm Reject.
-  Cancel.

Validations:

- If “**Other**” is selected → Notes field becomes mandatory.
- At least one reason must be selected.

→ Backend Update on Reject

When rejection is confirmed:

- Listing status changes from **Pending** or **Active** → **Rejected**.
- Store:
 - Reject reason.
 - Notes.
 - Admin ID who rejected it.
 - Date & time.

Audit Trail Entry:

“Listing [Property ID] rejected **by Admin [Admin ID] on [Date/Time] – Reason: [Reason/Notes]**”

Notifications on Rejection

- Agent/Developer Notification (via platform + email + Telegram)

1.5 Referral System Management

Description

The **Referral System Management** module allows Admins to configure and manage the platform's **Affiliate Program** in two possible modes:

1. **Standard Affiliate Model** → Each affiliate earns commission only on their **direct referrals' purchases**.
2. **Chain Referral Model (2-Level MLM)** → Affiliates earn commission from **direct referrals (Level 1)** and **indirect referrals (Level 2)**.

The Admin can toggle between modes, define commission rules, review affiliate applications, manage payouts, and monitor fraud.

Navigation

- Sidebar → **Referral Management**
 - Sub-tabs:
 1. Affiliate Applications
 2. Active Affiliates
 3. Commission Rules
 4. Referral Tracking
 5. Payout Management
 6. Reports & Audits

Global Referral Program Controls

- **Fields & Options:**
 - Enable/Disable Affiliate Program (toggle).
 - Referral Mode (dropdown: Standard / Chain Referral).

- **Validations:**
 - Must confirm before disabling (warn: “All affiliate tracking will stop immediately”).

Navigation: Referral Management → Settings.

Affiliate Applications

Description

The **Affiliate Applications** sub-module allows Admins to **review, verify, and approve/reject affiliate sign-up requests**.

- When an external user (Agent, Client, Developer, Advertiser, or Partner) applies to join the affiliate program, their request appears here.
- Admin can validate details and approve or reject the applicant to become an **Active Affiliate**.
- Once approved, the system automatically **generates a unique referral code/link** and grants the applicant **access to their Affiliate Dashboard**.

Navigation

- Sidebar → Referral Management → Affiliate Applications
- OR from Dashboard widget → “Pending Affiliate Applications” → Redirects to this page.

Affiliate Applications Page (List View)

Purpose: Show all pending and processed affiliate requests.

Table Columns:

1. Application ID (auto-generated).
2. Full Name.
3. Email.
4. Phone Number.
5. Applied On (date & time).
6. Referral Type (Standard / Chain).
7. Status (Pending / Approved / Rejected).
8. Actions.

Filters (Top Panel):

- Status (Pending, Approved, Rejected).
- Date Range (Applied On).
- Referral Type (Standard / MLM).

Search Bar: To search any application by Name / Email / Application ID.

Buttons (Global):

- **Refresh List** (reload latest applications).

Row Actions (per application):

-  **View Application** – Opens detail view of the application.
-  **Approve** – Approves application directly (with confirmation).
-  **Reject** – Rejects application (with reason).

Affiliate Application Detail Page

Purpose: Allow Admin to review the full application before decision.

Sections:

A. Applicant Info

- Full Name (required).
- Email (required, unique).
- Phone Number (required, unique, country code validation).
- User Type (Agent / Developer / Advertiser / Client / Other).
- Country & City.
- Preferred Payout Method (ABA)

B. Uploaded Documents (if required)

- ID Proof (image/PDF).
- Business License (for companies).

C. Application Metadata

- Application ID.
- Applied On (date & time).
- Referral Program Type requested (Standard / MLM).
- Current Status.

Admin Action Buttons (at bottom):

1.  **Approve Application**
 - Generates **unique referral code/link** (e.g., **REF12345**).
 - Sets affiliate status = Active.

- Grants access to **Affiliate Dashboard**.
- Sends email notification:
 - *“Your affiliate application has been approved. Your referral code: [CODE]. You can now log in to your affiliate dashboard.”*

2. **Reject Application**

- Opens rejection modal with **required reason field**.
- Options: Incomplete Details / Invalid Documents / Duplicate Account / Other (text input required).
- Sends email notification:
 - *“Your affiliate application has been rejected. Reason: [Admin Reason]. Please update your details and reapply if eligible.”*

Validations

- Email must be unique (not linked to another affiliate).
- The phone must be a unique and valid format.
- At least one payout method must be provided.
- Documents (if required by system settings) must be uploaded before approval.

End-to-End Flow

1. User applies for Affiliate Program (via registration page).
2. Application appears in **Admin** → **Referral Management** → **Affiliate Applications**.
3. Admin filters & reviews pending applications.
4. Admin opens detail view → verifies applicant info & documents.

5. Admin decides:

-  Approve → system generates referral code + grants dashboard access.
-  Reject → system requires reason + notifies applicant.

6. Application moves to **Active Affiliates** (if approved) or stays archived as Rejected.

1.6 Promotions & Discount Engine

The **Promotions & Discount Engine** module enables Admins to **create, configure, manage, and analyze promotional discounts**. It provides flexible tools to set discount rules, target specific user segments, restrict usage, and monitor impact on revenue.

This module applies discounts to:

- Subscription plans.
- Transaction fees.
- Listing upgrades.
- Other platform services.

Targeting can be done by **user segments** (new users, existing users, churn users, region-specific) and controlled with **eligibility rules and redemption limits**.

Navigation to Promotions & Discount Engine

Admins can access this module via multiple routes:

- 1. Login → Dashboard**
 - Admin logs in and lands on Dashboard.
- 2. Access via Sidebar Menu**
 - Click “**Promotions**” → Redirects to **Promotions Dashboard**.
- 3. Access via Quick Actions on Promotions Dashboard**
 - “**Create Promotion**” button → Redirects to Add Promotion Form.

Promotions Dashboard (Landing Page)

Purpose: Overview of all promotions & discount campaigns with quick management options.

UI Components (Table):

- Columns:
 - Promotion ID (auto-generated).
 - Name.
 - Discount Type (Fixed Amount / Percentage).
 - Value (e.g., \$100 / 20%).
 - Applicable Plans/Services (Plans, Fees, Upgrades).
 - Eligible Users (New, Returning, Geographic Segment).
 - Redemption Limit (e.g., first 100 users).
 - Start Date.
 - End Date.
 - Status (Active, Scheduled, Expired, Inactive).
 - Actions (View, Edit, Deactivate, Delete).

Filters (Top Panel):

- Status (Active, Expired, Scheduled).
- User Segment (New Users, Returning Users).
- Date Range.
- Discount Type.

Buttons:

-  Create Promotion.
-  Export Promotions Report (CSV, Excel, PDF).

Search Bar to Search Promotion by Name/ID.

Navigation: Clicking a promotion → Promotion Detail Page.

Promotion Detail Page

Purpose: View details and performance of a single promotion.

Sections: Each section includes the fields/data related to the promotion.

1. **Promotion Info:** Name, Description, Discount Type, Value, Dates, Status.
2. **Eligibility Rules:** User Segment, Applicable Plans, Geographic Restrictions.
3. **Redemption Stats:**
 - Total Redemptions.
 - Remaining Redemptions (if limited).
 - Revenue Impact.
4. **Promo Codes (if applicable):** List of generated codes with usage count.

Admin Actions:

- Edit Promotion.
- Deactivate Promotion (stop it immediately).
- Delete Promotion (permanent).

Navigation: Back button → Promotions Dashboard.

Add/Edit Promotion (Form)

Clicking on the (+) plus button on the promotion dashboard page will allow the admin to add/create a new promotion. This will open a page with empty below mentioned fields.

Clicking on the edit button will allow the admin to edit the promotion details. It will open a page with the prefilled data.

Fields (with validation):

- Promotion Name (required, min 3 chars).
- Description (optional, max 500 chars).
- Discount Type (dropdown: Fixed Amount / Percentage).
- Discount Value (required, numeric >0).
- Applicable Plans/Services (checkboxes: Plan A, Plan B, Listing Fees, Featured Listings, etc.).
- Eligible Users (dropdown: New Users Only, Returning Users, All Users, Geographic Segment).
- Redemption Limit (optional, numeric >0).
- Start Date (required, must be today or future).
- End Date (required, must be after Start Date).
- Status (Active / Inactive / Scheduled).

Buttons:

-  Save Promotion - It will save the data.
-  Cancel - Will redirect to promotion dashboard.

Validations:

- Start Date < End Date.
- Discount Value cannot exceed 100% (if percentage).
- The Redemption Limit must be integer.

Navigation: Save → Redirects to Promotion Detail Page with success message.

Promo Code Management

Purpose: Manage unique promo codes for campaigns/partners.

Fields (Form):

- Code (auto-generated or manually entered, must be unique).
- Linked Promotion (dropdown of active promotions).
- Max Usage (required, numeric).
- Expiry Date (required, after today).
- Status (Active / Inactive).

UI Components (Table):

- Promo Code.
- Linked Promotion.
- Usage Count.
- Max Usage.
- Expiry Date.
- Status.
- Actions (Deactivate | Delete).

Validations:

- Promo Code must be unique (check in DB).
- Expiry Date > today.
- Max Usage > 0.

Advanced Promotion Controls

1. User Segmentation:

- By user type (New / Existing / Churn).
- By location (Region-based).
- By behavior (optional: High spenders, Inactive users).

2. Stackable Rules:

- Toggle whether multiple promotions can be applied together.
- If No → system auto-blocks applying 2 discounts in checkout.

3. Redemption Rules:

- Max redemption count (e.g., first 100 users).
- Per-user redemption (e.g., one-time use per account).

Promotion Analytics & Reporting

Dashboard Charts & Reports:

- Active vs Expired Promotions.
- Redemption Rate (per promotion).
- Financial Impact (Revenue gained vs Revenue lost).
- User Segmentation (Who redeemed more – New vs Returning vs Churn).

- Promo Code Usage breakdown.

Admin Actions:

- Export Report (CSV, Excel, PDF).
- Generate Custom Report (select promotion IDs + date range).

Navigation: Promotions Dashboard → Analytics Tab.

Security & Rules

- Only **Super Admins** can delete promotions.
- Finance/Admin roles can create, edit, deactivate.
- All actions logged in the audit **trail**.
- The system must validate **promo eligibility** at checkout.
- Expired/Inactive promotions auto-blocked by backend validation.

End-to-End Flow

1. Admin logs in → Dashboard → Sidebar → Promotions.
2. Lands on **Promotions Dashboard**.
3. Clicks **+** Create Promotion → Fill Form → Save → Promotion Detail Page.
4. Can **add promo codes** (for campaigns/partners).
5. Promotion goes Live (Active status).
6. The system enforces rules at checkout (eligibility, validity, stackability, redemption limits).
7. Admin monitors redemptions in **Analytics Tab**.
8. If needed → Admin edits, deactivates, or deletes promotion.
9. Reports exported for revenue analysis.

1.7 Admin Profile & Profile Management (Admin Module)

Description

The **Admin Profile & Profile Management module** allows each Admin user to:

- Manage their **personal information**.
- Update **account settings** such as password, email, and phone.
- Configure **preferences** like language, timezone, and notifications.
- Upload and manage their **profile photo**.
- View and manage their **login sessions & activity logs** for security.

This ensures every Admin has control over their account and activity transparency, while Super Admins retain higher-level control.

Navigation

1. Top Navigation (User Avatar Dropdown)

- Admin clicks their **profile picture/avatar** (top right corner).
- Dropdown options:
 - Profile
 - Settings
 - Change Password
 - Logout

2. Sidebar → System Settings → Profile

- Accessible via **System Settings > Profile Management**.

Admin Profile Page (View Mode)

Purpose: Displays Admin's profile info in a read-only view.

Sections:

1. Profile Info Card:

- Profile Photo (default avatar if not uploaded).
- Full Name.
- Role (Super Admin / Finance Admin / Content Admin).
- Email.
- Phone Number.
- Status (Active/Inactive).
- Last Login Date/Time.

2. Quick Actions:

- Edit Profile.
- Change Password.
- View Activity Logs.

Navigation: Clicking **Edit Profile** → Profile Edit Page.

Edit Profile Page

When accessing the **Edit Profile Page**, the Admin can:

- **Update Personal Information** → such as name, email, and phone number.
- **Upload or Change Profile Photo** → to personalize their account.
- **Modify Preferences** → including preferred language, timezone, and notification settings.
- **Save Changes with Validation** → ensuring data integrity (e.g., valid email format, phone number formatting).

Any edits made here are reflected immediately across the system and are logged in the audit trail for security and compliance.

Fields (with validation):

- Profile Photo (upload image, max 5MB, formats: JPG/PNG).
- Full Name (required, min 3 chars).
- Email (required, must be unique + valid email format).
- Phone Number (optional, numeric + country code validation).
- Language Preference (dropdown: English, Khmer, Chinese + Google Translate toggle).
- Timezone (dropdown, default auto-detected).

Buttons:

-  Save Changes - It will save the data changes made in the form.
-  Cancel - It will not save any data change and return to the profile detail page

Validations:

- Email must be unique in the system.
- The phone must be numeric and formatted properly.
- Name cannot exceed 50 chars.

Navigation: Save → Redirect back to Profile Page with success message.

Change Password Page**Fields:**

- Current Password (required).
- New Password (required, min 8 chars, must include 1 uppercase, 1 lowercase, 1 number, 1 special character).
- Confirm New Password (must match new password).

Buttons:

-  Update Password - It will update the old password.
-  Cancel - It will not record any change and password does not changes.

Validations:

- Current Password must match existing DB record.
- New password ≠ old password.
- Confirm Passwords must match.

Navigation: On success → redirect to Login (force re-login).

Notification Preferences (Optional Tab)

Fields (toggles):

- Email Notifications (On/Off).
- Telegram Notifications (On/Off, requires linked account).
- System Alerts (On/Off).

Navigation: Profile → Notification Preferences.

LogOut

The Logout Flow allows an Admin to securely exit the system. Logging out ensures:

- Active session is terminated.
- Authentication tokens/cookies are invalidated.
- Sensitive data is not left open on shared devices.

The Logout option is always available from the Admin Profile menu (top-right avatar).

Navigation

1. Admin logs in → Dashboard.
2. Top-right corner → Profile Avatar Dropdown.
3. Options in dropdown:
 - My Profile
 - Preferences
 - Help/Support
 - Logout
4. Admin clicks Logout → Logout flow initiates.

Logout Flow (Step by Step)

Step 1 – Click Logout

- Admin clicks the Logout button in the Profile menu.

Step 2 – Confirmation Prompt (Optional)

- Confirmation modal appears (can be enabled in settings).
 - “Are you sure you want to log out?”
 - Buttons: Yes, Logout / Cancel.
- If disabled → Logout is immediate.

Step 3 – Session Termination

- System clears:
 - Access token (JWT/session key).
 - Refresh token (if any).
 - Cached data in local/session storage.
- System invalidates session on backend.

Step 4 – Redirect

- User redirected to Login Page.
- Success message shown:
 - “You have been logged out successfully.”

Buttons & Actions

- Logout (Profile Menu) → Initiates flow.
 - Yes, Logout (Modal) → Confirms logout.
 - Cancel (Modal) → Returns to dashboard, no action taken.
-

Backend APIs (Examples)

- POST `/api/admin/logout`
 - Invalidates current session tokens.
- DELETE `/api/admin/session`
 - Removes active session from server.

Activity Logs & Session Management

Purpose: Track Admin activity for security.

UI Components:

- Table of Login Sessions:
 - Session ID.
 - Device/Browser.
 - IP Address.
 - Login Date/Time.
 - Logout Date/Time.

- Status (Active / Expired).

Admin Actions:

- Force Logout (end suspicious session).

Navigation: Profile → Activity Logs Tab.

End-to-End Flow

1. Admin logs in → clicks Profile Avatar.
2. Lands on **Profile Page** (view mode).
3. Can:
 - Edit Profile (update personal info).
 - Change Password (secure reset).
 - Manage Notifications.
 - View Activity Logs.
4. Changes are saved with validations & logged in the audit trail.
5. Password change → triggers forced logout and requires re-login.

1.8 User Role Management (RBAC – Role-Based Access Control)

The **User Role Management** module gives Admins the ability to **create, configure, and manage user roles** for both:

1. **Internal Team Members** → e.g., Marketing Manager, Finance Officer, Content Reviewer.
2. **External Stakeholders** → Clients, Agents, Lenders, Developers, Advertisers.

Admins can define **roles and their permissions** to ensure every user only has access to the sections of the system they need. This helps maintain **security, accountability, and workflow control**.

Navigation

- **Sidebar** → **Roles & Permissions**
 - Clicking this opens the **Roles Dashboard**.
- Available to **Super Admins only**.

Roles Dashboard (Landing Page)

Purpose: View and manage all existing roles.

UI Components (Table):

- Role ID (auto-generated).
- Role Name.
- Role Type (Internal / External).
- Description.
- Number of Users Assigned.
- Status (Active / Inactive).
- Actions (View | Edit | Delete | Deactivate).

Filters (Top Panel):

- Role Type (Internal / External).
- Status (Active / Inactive).
- Search by Role Name.

Search Bar to Search Role by Name/ID.

Buttons:

-  Create New Role - This will open a page where user can create a new role with all the required fields.

Navigation: Click Role Name → Role Detail Page.

Create/Edit Role Page

This page opens to a page where users can create or edit roles.

When a user creates a role all the fields are empty in the page.

When a user wants to edit a role all the fields are prefilled in the page.

Fields (with validation):

1. Basic Info

- Role Name (required, unique, min 3 chars).
- Description (optional, max 300 chars).
- Role Type (dropdown: Internal / External).

2. Permission Matrix

- Display modules as a **list with checkboxes/toggles**:
 - Dashboard (View / No Access).
 - User Management (View / Edit / Delete).
 - Property Listings (View / Approve / Reject / Edit / Delete).
 - Transactions & Commissions (View / Manage / Export).
 - Promotions & Discounts (Create / Edit / Deactivate / Delete).
 - Reports & Analytics (View / Export).

- Settings (Full / Limited / None).

- Toggle for “**Full Access**” to grant all permissions in one click.

3. Status

- Active / Inactive.

Buttons:

- **Save Role** - It will save/record any change made by the user and return to the role dashboard page.
- **Cancel** - It will not record any changes in the data and will return to the role dashboard page.

Validations:

- Role Name must be unique.
- At least 1 permission must be selected.

Navigation: Save → Redirects to Role Detail Page.

Role Detail Page

Purpose: View complete information about a role.

Sections: Each section includes fields/data related to the role.

1. Role Information: Name, Type, Description, Status.
2. Assigned Permissions: List of all allowed actions.
3. Users Assigned: Table with User ID, Name, Email, Status.

Admin Actions:

- Edit Role.
- Deactivate Role (users lose access immediately).
- Delete Role (only if no users assigned).

Assign Roles to Users

Admins can assign roles when creating/editing a user.

Fields (in User Form):

- Role (dropdown list of available roles).
- Permissions preview (read-only display).

Navigation: User Management → Create/Edit User → Assign Role.

End-to-End Flow

1. Super Admin logs in → Sidebar → **Roles & Permissions**.
2. Lands on **Roles Dashboard**.
3. Can filter, search existing roles.
4. Clicks **Create New Role** → opens form.
5. Enters details + selects permissions via matrix.
6. Saves role → redirected to Role Detail Page.
7. While creating/editing a user, assigns them to the new role.

8. The user sees only the modules/actions allowed by their role.
9. Admin can later **edit, deactivate, or delete roles**.

1.9 System Settings & Configurations (Admin Module)

The **System Settings & Configurations** module allows Admins to **customize and manage platform-wide settings**, ensuring flexibility, personalization, and integration control.

It provides tools for:

- General system setup (branding, logo, languages, timezone).
- Security configurations (password rules, 2FA, session management).
- Payment & billing integrations.
- Communication and notifications setup.
- Third-party service integrations (CRM, Calendly, DocuSign, Telegram, Maps, etc.).
- Data, audit logs, and backup configurations.

This ensures the system is adaptable for business needs while keeping security, compliance, and user experience at the core.

Navigation

- **Sidebar → System Settings**
 - Expands into multiple tabs/sub-sections.

General Settings

Purpose: Control identity, and platform basics.

Fields:

- Platform Name (required).
- Default Language (English, Khmer, Chinese).
- Enable Google Translate (toggle).
- Default Timezone (dropdown).
- Contact Email & Phone.

Theme Modes

- Light Mode (default).
- Dark Mode.
- Auto Mode (switches based on system/browser setting).

Buttons:

- Save Settings.

Validations:

- Required fields must not be blank.

Navigation: System Settings → General.

Security Settings

Purpose: Manage account security and platform policies.

Fields & Options:

- Password Policy (min length, special chars, expiration days).
- Two-Factor Authentication (toggle, email/phone/Telegram).
- Session Timeout (e.g., auto logout after 30 mins).
- Concurrent Login Restriction (toggle: allow/disallow multiple logins).

Buttons:

- Save Security Settings.

Navigation: System Settings → Security.

Payment Gateway Configuration

Purpose: Manage payment integrations for transactions.

Options:

- Select Gateway:
 - ABA Payment Gateway.
 - Bakong.
- Fields:
 - API Key.
 - Merchant ID.
 - Secret Key.
 - Callback URL.
- Test Mode / Live Mode (toggle).

Buttons:

- Save Payment Configurations.
- Test Connection.

Validations:

- API fields must not be empty.
- Test connection must succeed before saving live mode.

Navigation: System Settings → Payments.

1.10 Reporting & Analytics (Admin Panel Module)

The **Reporting & Analytics** module gives Admins the ability to **monitor, analyze, and export key performance metrics** across the platform. It consolidates data from **Users, Agents, Affiliates, Promotions, and Revenue** into interactive reports.

Admins can:

- View pre-built reports (growth, sales, revenue, usage).
- Apply **filters and date ranges**.
- Export reports in **CSV, Excel, PDF**.

Navigation Flow

1. Admin logs in → Sidebar.
2. Click → **Reports & Analytics**.
3. Lands on **Reports Dashboard** with **KPIs Snapshot Widgets** (overview).
4. Tabs available at the top:

- User Growth
- Agent Performance
- Affiliate Sales
- Promotion Usage
- Revenue Reports

Reports

1. User Growth Report

Description: Shows how the platform's user base is expanding, segmented by type and activity.

Widgets/Graphs:

- Total Registered Users (big number card).
- Growth Trend (Line Chart: new users over time).
- User Type Breakdown (Pie Chart: Clients vs Agents vs Advertisers vs Developers).
- Active vs Inactive Users (Bar Graph).

Data Table Columns:

- User ID
- Name
- Email
- Role (Client / Agent / Advertiser / Developer)
- Signup Date
- Last Active Date
- Status (Active / Inactive)

Filters:

- Date Range (Today, 7 days, 30 days, Custom).
- User Type (multi-select).
- Location (Country / City).
- Status.

Buttons:

- Apply Filters
- Export Report (CSV, Excel, PDF)

Example Insight:

“200 new agents signed up in Phnom Penh in the last 30 days, with 75% being active weekly.”

2. Agent Performance Report

Description: Evaluates agents' contribution to listings, deals, and client satisfaction.

Widgets/Graphs:

- Total Active Agents.
- Top 5 Agents by Sales Volume (Leaderboard).
- Average Response Time (Bar Graph: per agent).
- Revenue Contribution by Agents (Stacked Chart).
- Client Rating Distribution (Pie Chart).

Data Table Columns:

- Agent ID

- Name
- Listings Count
- Sales Closed
- Revenue Generated
- Avg. Response Time
- Avg. Rating
- Status (Active / Suspended)

Filters:

- Date Range
- Location
- Property Category (Sale, Rent, Foreclosure, etc.)
- Performance Threshold (e.g., Sales > X)

Actions:

- View Agent Profile → Opens detail view.
- Export Performance Report.

Example Insight:

“Agent A closed 15 sales worth \$200,000 in the past quarter with a 4.8-star average rating.”

3. Affiliate Sales Report

Description: Tracks performance of affiliates and their referral-driven transactions.

Widgets/Graphs:

- Total Active Affiliates.
- Referrals Generated (Direct & Indirect).
- Conversion Rate % (Signups → Sales).
- Commission Distribution (Donut Chart).
- Top Affiliates by Sales (Leaderboard).

Data Table Columns:

- Affiliate ID
- Name
- Referral Code
- Direct Referrals
- Indirect Referrals (if MLM enabled)
- Conversions
- Commission Earned
- Payout Status (Pending / Paid)

Filters:

- Date Range
- Affiliate Type (Standard / MLM).
- Status (Active / Suspended / Blacklisted).

Actions:

- View Affiliate Profile (shows referrals, hierarchy, commissions).
- Export Affiliate Sales Report.

Example Insight:

“Affiliate John Doe generated 120 signups, 40 conversions, earning \$2,500 in commissions last month.”

4. Promotion Usage Report

Description: Analyzes how discount codes and promotions impact user behavior and revenue.

Widgets/Graphs:

- Total Promotions Created.
- Redemptions Count.
- Active vs Expired Promotions.
- Top 5 Promo Codes by Usage.

Data Table Columns:

- Promo ID
- Promo Name
- Type (Fixed / % Discount)
- Validity Period
- Target Segment (New Users, Returning Users, Agents, etc.)
- Redemptions Count
- Conversion Lift %
- Status (Active / Expired / Disabled)

Filters:

- Date Range
- Status (Active, Expired)
- Segment (New, Returning, Agents, Developers)

Actions:

- View Promo Details.
- Export Promotion Report.

Example Insight:

“Promo ‘WELCOME20’ was redeemed 320 times in 2 weeks, generating \$12,000 in sales uplift.”

5. Revenue Reports

Description: Financial breakdown of platform income and costs.

Widgets/Graphs:

- Total Revenue (Number Card).
- Revenue Streams (Pie Chart: Listing Fees, Subscription, Transaction Fees, Promotions).
- Monthly Revenue Trend (Line Chart).
- Net Profit after Costs (Bar Graph).
- Payout Costs (Commissions to Affiliates).

Data Table Columns:

- Transaction ID
- Date
- User Type
- Source (Listing Fee / Subscription / Promotion / Other)
- Amount (Debit / Credit)
- Net Revenue Impact

Filters:

- Date Range
- Source Type
- Region

Actions:

- Export Revenue Report.
- Compare Periods (e.g., This Quarter vs Last Quarter).

Example Insight:

“Revenue grew 15% this quarter, with subscription plans contributing 60% of total income.”